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Federal Wage System Series

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CAREER FIELD EDUCATION AND TRAINING PLAN

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Part I

Preface

This Career Field Education and Training Plan (CFETP) is a comprehensive education and training document that identifies life-cycle education/training requirements and training support resources for the identified series. The CFETP provides Federal Wage System (FWS) personnel with a clear career path to success and instill rigor in all aspects of our job series training.

The CFETP consists of two parts used by the supervisor to plan, manage, and control training within the job series.

Part I includes the following:

- **Section A** provides general information about how the CFETP will be used.
- **Section B** identifies job series progression information, duties and responsibilities, training strategies, and the job series path.

Part II includes the following:

- **Section A** identifies the Family Series Training Standard (FSTS) to include duties, tasks, and technical references to support FWS training programs.
- **Section B** identifies available support materials.
- **Section C** identifies a training course index supervisors can use to determine resources available to support training. Included here are both mandatory and optional courses, and exportable courseware.

Using guidance provided in the CFETP will ensure individuals in the identified series receive effective and efficient training at the appropriate point in their careers. This plan will enable us to train today's workforce for tomorrow's jobs. At the unit level, supervisors and trainers must use Part II to identify, plan, and conduct training commensurate with the overall goals of this guide and the local mission.

Section A: General Information

A1. Purpose

The CFETP may be used to document training and proficiency of the employee on associated task(s) by the supervisor or certified trainer. The FSTS (Attachment 2) portion of the CFETP is designed to be a tool for supervisors to use in assessing the skill level of current and new employees. The CFETP may be used to document training and proficiency of the employee on associated task(s) by the supervisor or certified trainer.

A2. Uses

Managers and supervisors may use the plan at all levels to ensure comprehensive and cohesive training programs are available for everyone in the FWS series.

A2.1. Federal Wage System Panel of the Functional Advisory Council

The Federal Wage System Panel of the Functional Advisory Council will develop/revise formal resident, non-resident, field, and exportable training based on requirements established by the users and documented in Part II of the CFETP. They will also work with the Air Force Civil Engineer Center Force Development Division (AFCEC/COF) to develop acquisition strategies for obtaining resources needed to provide the identified training.

A2.2. Federal Wage System Panel along with the Readiness and Training Managers

The Federal Wage System Panel along with the Readiness and Training Managers, will ensure their training programs complement the CFETP training requirements and identify requirements that can be satisfied by OJT, resident training, contract training, or exportable courses.

A2.3. Supervisors

Supervisors will guide individuals through completion of training specified in this plan.

A2.4. Individuals

Individuals complete training requirements specified in this plan. The list of courses in Part II of this CFETP will be used as a reference to support training.

A3. Coordination and Approval

The Air Force Civil Engineer Civilian Career Field Manager is the approval authority for the CFETP. The Federal Wage System Panel along with the Readiness and Training Managers will identify and coordinate on FWS series training requirements. Using the list of courses in Part II, they will eliminate duplicate training.

A4. Head Quarters United States Air Force (HQ USAF)/A4C

HQ USAF/A4C will review this CFETP annually and make updates and changes as deemed appropriate. Please send recommended changes to the AFCEC/COF Training Support Section at DSN 523-6879 or comm. 850-283-6879 or email.

Section B: FWS Progression and Information

B1. Series Descriptions

See specific Standard Core Personnel Document (SCPD) for the description.

B2. Skill and Career Progression

It is essential everyone involved in training do their part to participate in, plan, manage, and conduct effective training. The guidance provided in this part of the CFETP will identify viable training at appropriate points in an individual's career.

B2.1. Apprentice/Helper (A/H)

B2.1.1. Upon completion of initial skills training, an employee may work with a trainer to enhance their knowledge and skills to perform at the highest attainable level within their series.

B2.1.2. Utilize the CDC and other exportable courses for subject and task fundamentals in the series.

B2.1.3. Encourage apprentice/helpers to continue academic education and begin Enlisted Professional Military Education **(**EPME) by enrolling in Airman Leadership School either in-residence or by correspondence course.

B2.2. Journeyman (J)

B2.2.1. Journeymen may enhance their skills by engaging in additional training. After completing training, they may be assigned job positions such as team leader, trainer, or task certifier. Journeymen can pursue leadership training and skills to qualify for potential advancement to Work Leader or Work Supervisor positions.

B2.2.2. Encourage journeyman to enroll in the Noncommissioned Officer Academy (NCOA) either in-residence or by correspondence course.

B2.3. Craftsman (C)

B2.3.1. Craftsmen may continue to advance their skills by completing additional training. They may be assigned job positions such as team leader, trainer, or task certifier. Craftsmen are encouraged to pursue leadership training and skills to qualify for potential advancement to Work Leader or Work Supervisor positions.

B2.3.2. Encourage craftsmen to continue academic education and complete NCOA either in-residence or by correspondence course, civilian leadership courses and degree programs.

B2.3.3. Master Craftsman are typically graded higher than WG-10 where skills, knowledge and abilities require higher technical abilities than standard craftsmen. They are duty/location specific and not for all job series.

B2.4. Work Leader (WL)

B2.4.1. Work Leaders are expected to perform limited supervisory duties or act as a team lead.

B2.4.2. Completion of the Air Force Institute of Technology (AFIT) Civilian Supervisors Course (WMGT 571) is highly encouraged.

B2.4.3. Should pursue increased knowledge of budget, manpower, resources, and personnel management.

B2.4.4. Recommend pursuit of additional higher education and completion of courses outside of their job series for career broadening opportunities.

B2.4.5. Encourage Work Leaders to continue academic education and complete NCOA either in-residence or by correspondence course, Emerging Leader course and degree programs.

B2.5. Work Supervisor

B2.5.1. A supervisor can be expected to fill positions such as the Element Chief or Special Projects Supervisor.

B2.5.2. Must enroll and complete required mandatory supervisor or manager training courses within 1 year of appointment to a supervisory or managerial position and complete experience training every 3 years, thereafter.

B2.5.3. Completion of the AFIT Civilian Supervisors Course (WMGT 571) is highly encouraged.

B2.5.4. Should pursue increased knowledge of budget, manpower, resources, and personnel management.

B2.5.5. Recommend pursuit of additional higher education and completion of courses outside of their job series for career broadening opportunities.

B2.5.6. Encourage supervisors to continue academic education and complete Senior Noncommissioned Officer Academy (SNCOA) by correspondence, civilian leadership courses and degree programs.

B3. Correspondence Course Directions

Nonresident attendance for professional military education courses is accomplished through the Air Force (AF) Portal.

B3.1. Step 1

Login to the AF Portal.

B3.2. Step 2

Copy and paste the Air University URL into your browser.

B3.3. Step 3

Choose the “Distance Learning” tab on the on the right-side menu.

B3.4. Step 4

Select the appropriate course.

B4. Enlisted Professional Military Education (EPME)

Scheduling enlisted professional military education for civilian personnel is a responsibility of AFPC/DP3DW unless otherwise noted. See Department of Air Force Instruction (DAFI) 36-2670, *Total Force Development*, for more detailed information.

B4.1. Airman Leadership School (ALS)

Airman Leadership School resident attendance is scheduled by the local ALS commandant. Commandants build an annual schedule shortly after the staff sergeant (E-5) promotion release. Eligible AF and Department of Defense (DoD) civilians are considered priority 3 for ALS. DAFI 36-2670 encourages ALS leadership to the extent possible, distribute Priority 2 and 3 students throughout the year to maximize diversity in the classroom.

B4.2. Noncommissioned Officer Academy (NCOA)

Selection and scheduling are accomplished by AFPC/DPZ. Eligible AF and DoD civilians are considered priority 3 for NCOA and are considered on a space available basis.

B4.3. Senior Noncommissioned Officer Academy (SNCOA)

Selection and scheduling are accomplished by AFPC/DPZ and is driven by the senior master sergeant (E-8) promotion release. Eligible AF and DoD civilians are considered priority 4 for SNCOA and are considered on a space available basis.

B5. Career Building Blocks (CBBs) and Continuous Development Framework Model

B5.1. Career Building Blocks (CBBs)

The CBBs illustrate the dispersion of grades and relative experience levels (apprentice, journeyman, craftsman). As you progress through the grades, so should your breadth of experience. This not only includes technical experience, but leadership, supervisory and management experience as well. Using various on-the-job training, military, and civilian vocational schools, and professional military education (PME), forms the foundation of the CBBs and the "Develop Exceptional Leaders" portion of the strategic vision in the Civil Engineer (CE) Human Capital Road Map.

B5.1.1. CBBs

The CBBs are an illustration or an example of your individual development plan or career path. Your individual roadmap may be different than others based on your occupational series and/or career path. The CBBs provide a few examples of development opportunities at the tactical, operational, and strategic levels of performance. Click here to view the CBBs for your job series.

B5.1.2. Career Goals

Once you have established your career goals, identify a mentor to help you align your steps in this learning and development continuum. A mentor can help you identify different kinds of experience and training you will need as well as the tools to attain them. Mentors also serve as a sounding board and can help you reassess or adjust your career goals when events in your life or career necessitate changes. As you complete these steps and move toward your career goals, it is important to reassess your goals. Once this is accomplished, you and your mentor will be able to further your progression and advancement through the continuous development framework model. Refer to Figure 1 (below) for visual reference.

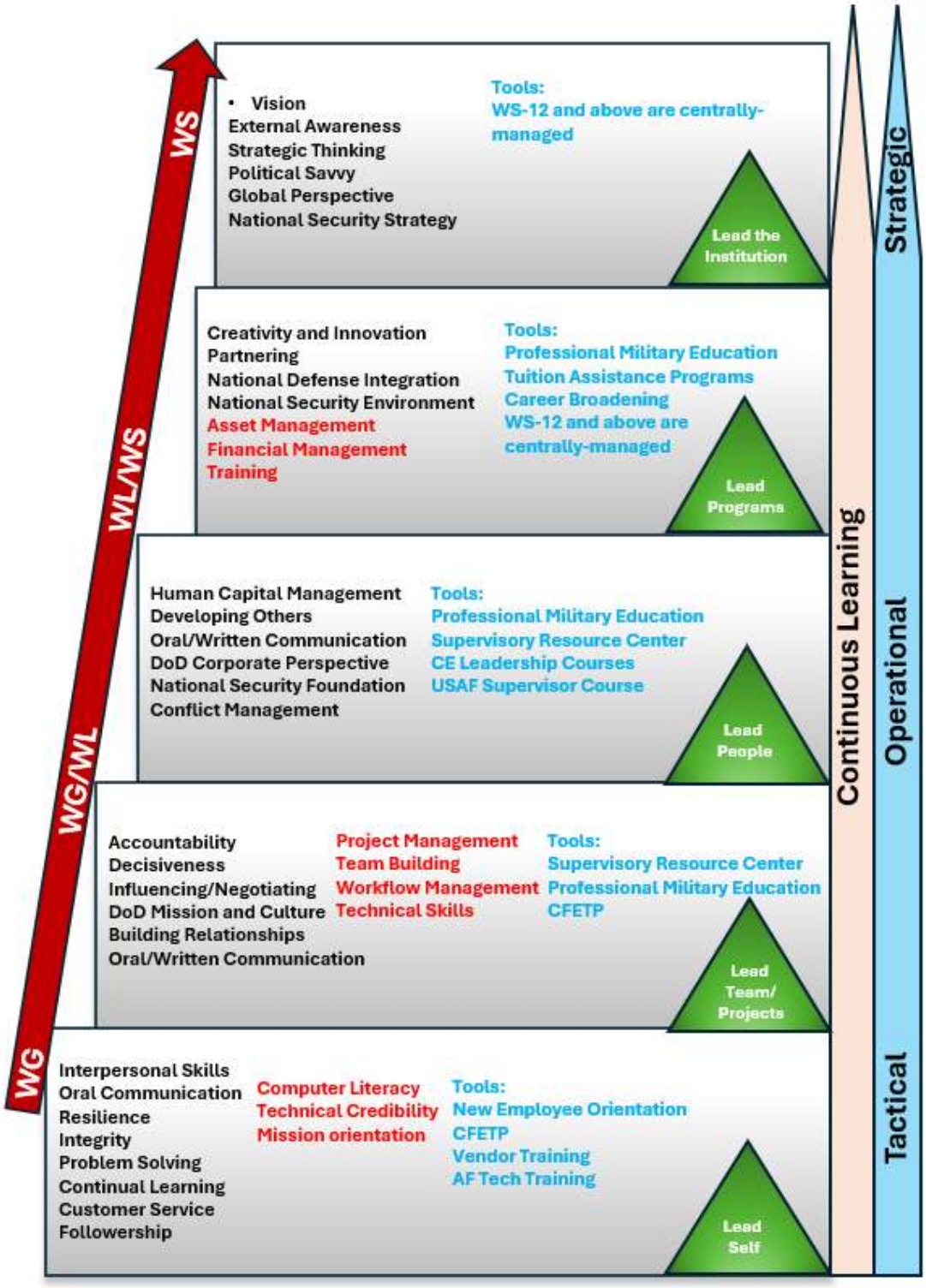


Figure 1: Continuous Development Framework Model

Part II

Section A: Family Series Training Standard (FSTS)

A1. Implementation

This FSTS is used for technical training provided by Air Education and Training Command (AETC), OJT and COTS training documentation.

A2. Purpose

As prescribed in DAFI 36-2670, *Total Force Development*, DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development* and in collaboration with the Air Force Civilian Career Field Manager (AFCCFM), this FSTS is mandatory for all FWS job series working under the AF Specialty Code 3E1X1, regardless of duty assignment. Each employee must use an automated training record.

A2.1. Column 1 (*Tasks, Knowledge, and Technical References*)

Lists the most common performance and knowledge requirements and supporting technical references (TR) necessary for an employee to perform duties in the Apprentice, Journeyman, Craftsman, and Supervisor level successfully. The required behaviors will be used as the context for which learning will be assessed.

A2.2. Column 2 (Core Tasks)

The CFM or shop foreman is responsible for identifying the mandatory tasks for each duty position within a job series at their specific location. Tasks identified have Core and/or Certification requirements, Civilian Deployment needs, or those designated with a Special Experience Identifier (SEI).

A2.2.1 Column 2 (*Deployment/SEI*)

Tasks identified as mandatory for employees who perform roles in Mission Critical, Mission Essential or have been identified for a civilian deployment tasking.

A2.3. Column 3 (Provides Certification for OJT)

Used to record completion of tasks and knowledge training requirements. Task certification requires the task to be trained by a trainer designated by the supervisor. The trainer can be either civilian or military. The training start and completion date are documented, the task is signed by the employee and either the work center supervisor, a Staff Sergeant (or above), Civilian equivalent, or the unit training manager. This action will complete the task certification.

Note: The "trainer" signing the record MUST be the work center supervisor, work leader, a Staff Sergeant (or above), Civilian equivalent, or the Unit Training Manager. This person does not necessarily train the task but will ensure the training is conducted by a qualified trainer prior to completing task certification.

Note: If a work center supervisor, Work Leader, a Staff Sergeant (or above), Civilian equivalent, or the unit training manager are not available in a shop or unit to certify a task, the Operations Flight deputy commander will designate a certifier within the flight.

A2.3.1. Task Certification

Task certification of core and critical tasks. Require a training completion date and initials of the employee, trainer, and a certifier. All non-core tasks require training completion date and initials of the employee and trainer only.

A2.3.2. Performance Standard

All training requirements are trained and qualified to the "Go" level. "Go" means the individual can perform the task without assistance and meets local demands for accuracy, timeliness and, if applicable, correct use of procedures and Technical Orders.

A2.4. Column 4 (Codes Used to Indicate level of Training)

Indicates whether the task is a Knowledge (K), Performance (P) or Performance and Knowledge (PK). Codes are provided in columns labeled for each level of experience. These are labeled A/H for Apprentice/Helper, J for Journeyman, C/WL for Craftsman or Work Leader and S for Supervisor. Identifies duty position tasks and their associated training requirements using a proficiency code. It shows the proficiency to be demonstrated on the job by the employee as a result of hands-on training on the task, knowledge and the career knowledge provided by formal courses, career development courses (CDCs), distance learning (DL) web-based training (WBT) and Air Force Qualification Training Packages (AFQTPs). CDC listings are maintained by the unit training manager (UTM) for current CDC listings.

A2.5. Qualitative Requirements.

Contains the proficiency code key used to indicate the level of training and knowledge provided by computer-based training (CBT), WBT, commercial off the shelf (COTS), in-resident training, and career development courses.

A2.6. Job Qualification Standard (*JQS*)

The FSTS becomes the Job Qualification Standard (JQS) for on-the-job training (OJT) when entries are made in the FSTS. For OJT, the tasks in Column 1 are trained and qualified to the go/no go level. "Go" means the individual can perform the task without assistance and meets local requirements for accuracy, timeliness, and correct use. JQS is used according to DAFI 36-2670, *Total Force Development* and DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development*. AFQTPs are available on myLearning to ensure that all supervisors use standardized procedures for training. When used as a JQS, the following requirements listed below apply.

A2.6.1. Documentation

Document and certify completion of training.

A2.6.1.1. Duty position

Requirements for each duty position (task group) will be developed and identified by the work center supervisor and loaded into the automated training

management application. Completion of the identified tasks is mandatory for all duty positions. Ensure the correct duty position title is listed in the Profile section of the employee's automated training record.

A2.6.1.2. AFQTP Training and Documentation

AFQTPs have been created for several task groups to fulfill performance (P) and knowledge (K) requirements for upgrade/qualification training. Each AFQTP provides the step–by–step procedures for the trainee, trainer, and certifier in completing each.

A2.6.1.2.2. Hands-On Training

For performance (P) training requirements, DO NOT sign off the tasks in the JQS until the employee has completed hands-on/ certification training.

A2.6.2. Transcribing from previous versions to new CFETP

The UTM and supervisor must conduct a review of the new STS to identify any new tasks and add those tasks to their unit specific duty positions.

A2.6.2.1. Previous training certification not listed

If previous training certifications are not listed in the individual training record, select the task to be transcribed, and click on the transcribe button. Enter the date of the original certification and sign off the task(s). The employee will then sign off the task(s) to finalize the transcription of previous training certification. The automated application will place an entry into the employee's training record and must be acknowledged by the transcriber and employee.

A2.6.2.2. Transcribing external training certification

If an employee attended a formal training course and received appropriate accreditation, select the formal training section of the users automated training record and locate the course title in the master task list, then enter the completion date. If the course is not listed, contact the Unit Education Training Manager (UETM) to have it loaded from the master catalog. If it is not listed in the master catalog, contact AFCEC/COF to have it loaded in the master catalog.

Section B: Course Objective List

B1. Measurement

Measurement of each learning objective is indicated as follows:

B1.1. Use of Progress Checks (PCs) & Rubrics

Indicates formal measurement of knowledge (K) and/or performance (P) elements.

B2. Standard

Standards for measurement are indicated in the course objectives and delineated on the individual progress checklist and rubrics. The minimum standard is 70% on knowledge progress checks. Trainer assistance is used as the standard for performance progress checks and is provided, as warranted during the progress check. Employee may be required to repeat all or parts of the learning outcome until satisfactory performance is attained.

B3. Proficiency Level

Student must demonstrate mastery on each learning outcome/objective before progressing to the next learning requirement.

B4. Course Objective List

These objectives are listed in the sequence taught by Blocks of Instruction. Per Air Education and Training Command Instruction (AETCI) 36 - 2651, *Basic Military and Technical Training*, a detailed listing of the initial skills course learning objectives in the Basic Course are listed in the 3E4X1 Air Force Specialty Code (AFSC) STS.

Section C: Support Material

C1. Air Force Qualification Training Packages (AFQTP)

C1.1. AFQTP Documentation Record

The AFQTPs for each task group are identified on the AFQTP Documentation Record located in Attachment 3.

C1.2.1. Applicable AFQTPs

For a complete list of up-to-date applicable AFQTPs go to myLearning.

C1.2.2. Web-based Courses

In addition to the AFQTPs, there are web-based courses or assessments developed for certain tasks that are available on myLearning under AFCEC in the specialty topic area.

C2. Career Development Course (CDC) Assessment for CE CDC/DL Course

C2.1. CDC Assessments

AFCEC/COF has developed CDC assessments for each AF CE career field, and they are located on the myLearning under AFCEC in the topic header CE CDCs Assessments.

C2.2. CDC Purpose

CDC assessments are for the sole purpose of providing the trainer and the supervisor a predictive indicator of whether the employee has a solid grasp on the knowledge portions of the STS.

Section D: Education and Training Course Index

D1. Purpose

This section of the CFETP identifies training courses available for the 3E1X1 Specialty. Refer to Education and Training Course Announcements for information on the AF in-residence courses.

D2. Air Force In-Residence Courses/Mobile Training Team (MTT) Courses

Course Number	Title	Developer
J8AQR3E031 01AB	HVAC/R Apprentice ITRO Qual 1	366 TRS
J3ABR3E131 03AC	HVAC/R Apprentice AF Unique	366 TRS
J3ACR3E171 00AC	HVAC/R Craftsman Course	366 TRS
J3AZR3E151 00BA	Advanced HVAC/R Air and Hydronic Balancing	366 TRS
J3AZR3E151 00AC	Advanced HVAR/R Building Automation Systems	366 TRS

Table 1: AF In-Residence Courses/MTT Courses

D3. Air Force Career Development Academy (AFCDA) Courses

Course Number	Title	Edit Code
CDC 3E151	HVAC/R Journeyman	01

Table 2: AFCDA Courses

D4. Exportable/Web-based Courses/Information

Course Number	Title	Developer
Web based	Arc Flash Safety Awareness QTP	AFCEC/COF
Web based	Civil Engineer 5-Level Core Concepts Course	AFCEC/COF
Web based	Civil Engineer 7-Level Core Concepts Course	AFCEC/COF
Web based	Confined Space Course	AFCEC/COF
Web based	Industrial Water Treatment Orientation Program	AFCEC/COF
Web based	130K BTU Heater	AFCEC/COF
Web based	3E1X1 Paper Based AFQTPs	AFCEC/COF
Web based	TRICON Refrigerated Container System (TRCS)	AFCEC/COF
Web based	HVAC/R Basic Electrical Diagnostics	AFCEC/COF
WENG 170	Cybersecurity for Control Systems	AFIT
WENG 200	Scoping and Estimating Course	AFIT
WENG 270	Advanced Control System Cybersecurity Course	AFIT
WENG 370	Control Systems Cybersecurity for CE Leaders	AFIT
WENG 400	Life-cycle Cost Estimating	AFIT
WENG 460	Introduction to Mechanical Systems	AFIT
WENG 563	HVAC Control Systems	AFIT
WENV 531	Air Quality Management	AFIT
WMGT 131	SMS Builder Course	AFIT
WMGT 301	Intro to Asset Management	AFIT
WMGT 322	Introduction to Project Management	AFIT
WMGT 436	Requirements and Optimization	AFIT
WMGT 437	Troop Construction Project Management	AFIT
WMGT 570	Civil Engineer Superintendent Course	AFIT
WMGT 571	Operations Flight Civilian Supervisor Course	AFIT

Table 3: Exportable/Web-based Courses

D5. Leadership Training and Academic Fellowships through the Eaker Center

Title:

- Squadron Officers School (SOS)
- Developing Supervisor Course
- Defense Emerging Leader Program
- Civilian Associate's Degree Program
- Civilian Bachelor's Degree Program
- Civilian Master's Degree Program

Located at: Civilian Force Development Homepage

Attachment 1 - Qualitative Training Requirements (Proficiency Code Key)

A1. Qualitative Training Requirements

This Block is for Identification Purposes Only.		
Name Of Employee		
Printed Name (*Last, First, Middle Initial*)	**Initials (*Written*)**	**SSAN (*Last four*)**
Printed Name of Trainer, Certifying Official and Written Initials		
N/I	*N/I*	

Attachment 2 - FWS Family Series Training Standard (FSTS)

A2. Family Series Training Standard (FSTS)

A2.1. Identification

In the training record User Profile section, the UTM will assign individuals to the correct work center, upon in-processing into the unit.

A2.2. Specialty Tasks

The following are tasks the work center supervisor will use to track each duty position created for their work center.

Behavioral Statement FSTS Coding System	
Code	**Definition**
K	Subject Knowledge Training - The verb selection identifies the individual's ability to identify facts, state principles, analyze, or evaluate the subject.
P	Performance Training - Identifies that the individual has performed the task to the satisfaction of the trainer/certifier; however, the individual may not be capable of meeting the field requirements for speed and accuracy.
PK	Performance Knowledge Training - The verb selection identifies the individual's ability to relate advanced facts, procedures, operating principles, and operational theory for the task.
X	Formal Course
*	Core Task. These are mandatory tasks identified by the Career Field Manager

Table 4: Behavioral Statement FSTS Coding System

See the Federal building competency model

1. Tasks, Knowledge And Technical References	2. Tasks	3. Certification For OJT					4. Proficiency Codes			
	Core/Cert	A	B	C	D	E	A	B	C	D
		Tng Start	Tng Complete	Trainee Initials	Trainer Initials	Certifier Initials	A/H	J	C/WL	S
1. CIVIL ENGINEER (CE) CORE CONCEPTS and FORMAL TRAINING COURSES:										
1.1. Civil Engineer (CE) 5-Level Core Concepts Course							X	X	X	X
1.2. Civil Engineer (CE) 7-Level Core Concepts Course								X	X	X
1.3. WMGT 100 Air Force Civil Engineer Basic Course (On-demand web course)							X	X	X	X
1.4. WMGT 322 Introduction to Project Management Course (Self-paced on-demand web course)								X	X	X
1.5. Complete WMGT 131 SMS BUILDER Level 1							X	X	X	X
1.6. Complete AFIT WENG 170 Cybersecurity for Control Systems	*						X	X	X	X
1.7. Complete WMGT 231: SMS BUILDER Level 2									X	X
1.8. Complete AFIT WMGT 301 Intro to Asset Management							X	X	X	X
1.9. Complete AFIT WENG 270 Advanced Control Systems Cybersecurity Course	*						X	X	X	X
1.10. Complete AFIT WMGT 322 Project Management								X	X	X
1.11. Complete AFIT WENG 370 Control Systems Cybersecurity for CE Leaders	*								X	X
1.12. WENG 460 Introduction to Mechanical Systems (Self-pace web course)								X	X	X

1.13. WENG 563 HVAC Control Systems (Self-paced web course)								**X**	**X**	**X**
1.14. Civilian Supervisor Course									**X**	**X**
1.15. WMGT 571 Operations Flight Civilian Supervisor Course (In resident)									**X**	**X**
2. HVAC/R SAFETY: TR: AFOSH 91-501, NFPA 70E										
2.1. Arc Flash Safety							**K**	**K**	**K**	**K**
2.1.1. Remove victim from energized circuits							**PK**	**PK**	**PK**	**PK**
2.1.2. Apply first aid procedures for electrical shock							**PK**	**PK**	**PK**	**PK**
2.2.. Confined space training:										
2.2.1. Confined space entry supervisor									**PK**	**PK**
2.2.2. Confined space attendant							**P**	**PK**	**PK**	**PK**
2.2.3. Confined space entrant							**P**	**PK**	**PK**	**PK**
2.3. Fall protection training							**P**	**PK**	**PK**	**PK**
3. HVAC/R PUBLICATIONS: TR: Fundamentals of HVAC/R 2nd Ed AHRI										
3.1. Technical Orders and Manufacturer's Manuals							**K**	**K**	**K**	**K**
3.2. Data Plate Specifications							**K**	**K**	**K**	**K**
3.3. National/DoD Certification requirements								**K**	**K**	**K**
3.4. Unified Facilities Criteria (UFC)									**K**	**K**
4. TOOLS & TEST EQUIPMENT: TR: TO's 32,33,34,35,38 series; Fundamentals of HVAC/R 2nd Ed AHRI										
4.1. Maintain and use tools (hand and powered)							**PK**	**PK**	**PK**	**PK**
4.2. Maintain and use precision measuring instrument:										
4.2.1. Digital temp sensors							**P**	**PK**	**PK**	**PK**
4.2.2. Digital humidity sensor							**K**	**PK**	**PK**	**PK**
4.2.3. Pyrometers							**K**	**PK**	**PK**	**PK**
4.2.4. Anemometers/Flow hood							**K**	**PK**	**PK**	**PK**
4.2.5. Gas Analyzers							**K**	**PK**	**PK**	**PK**
4.2.6. Psychrometer							**K**	**PK**	**PK**	**PK**
4.3. Maintain and use electrical test equipment:										
4.3.1. Digital Multimeter							**P**	**PK**	**PK**	**PK**
4.3.3. Ammeter							**P**	**PK**	**PK**	**PK**
4.3.4. Signal Generators							**K**	**PK**	**PK**	**PK**

4.4. Maintain and use computer equipment:										
4.4.1. Laptop computer stations							**K**	**PK**	**PK**	**PK**
4.4.2. Perform equipment firmware updates							**K**	**PK**	**PK**	**PK**
4.4.3. Troubleshoot HVAC equipment using laptop computer							**K**	**PK**	**PK**	**PK**
5. PIPING/TUBING: TR: Fundamentals of HVAC/R 2nd Ed AHRI										
5.1. Types & sizes							**K**	**K**	**K**	**K**
5.2. Fittings							**K**	**K**	**K**	**K**
5.3. Piping systems fabrication:										
5.3.1. Fabricate piping and tubing systems							**K**	**PK**	**PK**	**PK**
5.3.2. Install piping and tubing systems							**K**	**PK**	**PK**	**PK**
5.4. Types and operation of valves							**K**	**K**	**K**	**K**
5.5. Interpret system drawings							**K**	**K**	**K**	**K**
6. BRAZING & SOLDERING: TR: ASM Handbook V-6 Welding Brazing and soldering										
6.1. Theory of operation:										
6.1.1. Acetylene equipment (air and hydrocarbon)							**K**	**K**	**K**	**K**
6.1.2. Oxyacetylene equipment							**K**	**K**	**K**	**K**
6.2. Use equipment:										
6.2.1. Cut							**P**	**PK**	**PK**	**PK**
6.2.2. Braze and solder							**P**	**PK**	**PK**	**PK**
6.3. Maintain equipment							**P**	**PK**	**PK**	**PK**
7. HVAC/R PHYSICS: TR: Fundamentals of HVAC/R 2nd Ed; AHRI Environmental Systems Technologies 2nd Ed; Principles of Refrigeration 5th Ed										
7.1. Structure of matter							**K**	**K**	**K**	**K**
7.2. Energy (Stored, Non-stored, Conversion)							**K**	**K**	**K**	**K**
7.3. Laws of thermodynamics							**K**	**K**	**K**	**K**
7.4. Heat flow (Energy, Measurement, Transfer)							**K**	**K**	**K**	**K**
7.5. Fluid flow (Properties, Statics, Dynamics)							**K**	**K**	**K**	**K**
7.6. Psychrometrics:										
7.6.1. Properties of air							**K**	**K**	**K**	**K**
7.6.2. Air-vapor relationship							**K**	**K**	**K**	**K**
7.6.3. Psychrometric chart							**K**	**K**	**K**	**K**
7.6.3.1. Terms							**K**	**K**	**K**	**K**

7.6.3.2. Psychrometric processes:										
7.6.3.2.1. Plot processes of conditioned air							**K**	**PK**	**PK**	**PK**
7.6.3.2.2. Interpretation of plotted processes							**K**	**PK**	**PK**	**PK**
7.7. Load Calculations:										
7.7.1. Calculate building heat load							**K**	**PK**	**PK**	**PK**
7.7.2. Determine system requirements							**K**	**PK**	**PK**	**PK**
8. AIR and HYDRONIC SYSTEMS: TR: Environmental Systems Technology 2nd Ed; Trane Air Conditioning Manual; Principles of Refrigeration 5th Ed.; Refrigeration and Air Conditioning 4th Ed (ARI)										
8.1. Air Systems:										
8.1.2. Principles of air distribution systems							**K**	**K**	**K**	**K**
8.1.3. Constant Volume							**K**	**K**	**K**	**K**
8.1.4. Variable Air Volume (VAV)							**K**	**K**	**K**	**K**
8.1.5. Duct airflow characteristics							**K**	**K**	**K**	**K**
8.1.6. Multi-zone							**K**	**K**	**K**	**K**
8.1.7. Dedicated Outside Air System							**K**	**K**	**K**	**K**
8.1.8. Fan and system curve relationships								**K**	**K**	**K**
8.1.9. Adjust dampers and linkages							**K**	**PK**	**PK**	**PK**
8.1.10. Perform air balancing							**K**	**PK**	**PK**	**PK**
8.1.11. System drawings							**K**	**K**	**K**	**K**
8.2. Hydronic Systems:										
8.2.1. Principles of hydronic distribution systems							**K**	**K**	**K**	**K**
8.2.2. Pumps:										
8.2.2.1. Types and features							**K**	**K**	**K**	**K**
8.2.2.2. Install circulating pumps							**P**	**PK**	**PK**	**PK**
8.2.2.3. Adjust centrifugal water pump flow							**K**	**PK**	**PK**	**PK**
8.2.2.4. Replace packing on water pumps							**K**	**PK**	**PK**	**PK**
8.2.2.5. Replace mechanical seals/couplings							**K**	**PK**	**PK**	**PK**
8.2.2.6. Pump and system curve relationships							**K**	**K**	**K**	**K**
8.2.3. Hydronic system flow:										
8.2.3.1. Hydronic volume measurement (methods, interpretation)								**PK**	**PK**	**PK**
8.2.3.2. Balance hydronic system								**PK**	**PK**	**PK**
8.2.4. Interpret system drawings								**K**	**K**	**K**

8.3. Associated Components and Equipment:										
8.3.1. Coils (Types & Applications):										
8.3.1.1. Cooling							**K**	**K**	**K**	**K**
8.3.1.2. Heating							**K**	**K**	**K**	**K**
8.3.1.3. Preheat							**K**	**K**	**K**	**K**
8.3.1.4. Reheat							**K**	**K**	**K**	**K**
8.3.2. Fan coil units							**K**	**K**	**K**	**K**
8.3.3. Filters							**K**	**K**	**K**	**K**
8.4. Install							**K**	**PK**	**PK**	**PK**
8.5. Troubleshoot and repair							**K**	**PK**	**PK**	**PK**
8.6. Perform Preventive Maintenance:										
8.6.1. Inspect and replace drive belts							**P**	**PK**	**PK**	**PK**
8.6.2. Adjust drive belt tension							**P**	**PK**	**PK**	**PK**
8.6.3. Adjust pulleys							**P**	**PK**	**PK**	**PK**
8.6.4. Clean strainers							**P**	**PK**	**PK**	**PK**
8.6.5. Clean air filters							**P**	**PK**	**PK**	**PK**
8.6.6. Inspect water valves for leaks							**P**	**PK**	**PK**	**PK**
8.6.7. Inspect fan coil units							**P**	**PK**	**PK**	**PK**
8.6.8. Clean air handlers							**P**	**PK**	**PK**	**PK**
8.6.9. Clean coils							**P**	**PK**	**PK**	**PK**
8.6.10. Service water pumps							**P**	**PK**	**PK**	**PK**
9. ELECTRICAL: **TR: ANSI Y32.2; 32-1064; National Electric Code (NFPA 70); National Electrical Safety Code Handbook; Fundamentals of HVAC/R 2nd Ed AHRI**										
9.1. Principles of electricity (AC and DC)							**K**	**K**	**K**	**K**
9.2. Types of circuits:										
9.2.1. Line Circuit							**K**	**K**	**K**	**K**
9.2.2. Load Circuit							**K**	**K**	**K**	**K**
9.2.3. Control Circuit (110/208volt) AC/DC)							**K**	**K**	**K**	**K**
9.3. Relationship of current, voltage, and resistance in circuits							**K**	**K**	**K**	**K**
9.4. Wiring Diagrams							**K**	**K**	**K**	**K**
9.5. Wiring Requirements:										
9.5.1. Supply Voltage							**K**	**K**	**K**	**K**
9.5.2. Wiring color code							**K**	**K**	**K**	**K**
9.5.3. Wire sizes							**K**	**K**	**K**	**K**
9.5.4. Distribution Panels								**K**	**K**	**K**

9.6. Protective devices:										
9.6.1. Circuit Breakers							**K**	**K**	**K**	**K**
9.6.2. Fuses							**K**	**K**	**K**	**K**
9.6.3. Ground Fault Current Interrupters (GFCI)							**K**	**K**	**K**	**K**
9.6.4. Motor Overload Protectors							**K**	**K**	**K**	**K**
9.6.4.1. Motor starters							**K**	**K**	**K**	**K**
9.6.4.2. Internal motor overload protection							**K**	**K**	**K**	**K**
9.7. Devices:										
9.7.1. Switches							**K**	**K**	**K**	**K**
9.7.2. Receptacles							**K**	**K**	**K**	**K**
9.7.3. Timers							**K**	**K**	**K**	**K**
9.7.4. Transformers							**K**	**K**	**K**	**K**
9.7.5. Types and principles of electromagnetic devices (relays, contactors, and across-the-line starters)								**K**	**K**	**K**
9.8. Motors:										
9.8.1. Single phase and three phase							**K**	**K**	**K**	**K**
9.8.2. Replace motors							**P**	**PK**	**PK**	**PK**
9.8.3. Perform operational test							**K**	**PK**	**PK**	**PK**
9.8.4. Align pulleys/couplings							**P**	**PK**	**PK**	**PK**
9.8.5. Electrically connect (single-phase and three phase)							**K**	**PK**	**PK**	**PK**
9.8.6. Reverse rotation of electric motors							**K**	**PK**	**PK**	**PK**
9.8.7. Measure motor current draw							**K**	**PK**	**PK**	**PK**
9.8.8. Service electrical motors								**PK**	**PK**	**PK**
9.9. Types and principles of motor controllers and variable frequency drives								**K**	**K**	**K**
9.10. Install electrical components							**K**	**PK**	**PK**	**PK**
9.11. Troubleshoot and repair electrical circuits and components							**K**	**PK**	**PK**	**PK**
9.12. Perform preventive maintenance on electrical circuits and components							**K**	**PK**	**PK**	**PK**

10. HVAC CONTROL SYSTEMS: TR: AFMAN 32-1093; Fundamentals of HVAC/R 2nd Ed AHRI										
10.1. Types:										
10.1.1. Pneumatic HVAC Control Systems							**K**	**K**	**K**	**K**
10.1.2. Electrical HVAC Control Systems							**K**	**K**	**K**	**K**
10.1.3. Electronic HVAC Control Systems							**K**	**K**	**K**	**K**
10.1.4. Constant Air Volume (CAV)							**K**	**K**	**K**	**K**
10.1.5. Variable Air Volume (VAV)							**K**	**K**	**K**	**K**
10.1.6. Direct Digital Control (DDC)							**K**	**K**	**K**	**K**
10.2. Components:										
10.2.1. Variable Frequency Drives (VFD)							**K**	**K**	**K**	**K**
10.2.2. Principles sequence of operation							**K**	**K**	**K**	**K**
10.3. Calibrate and adjust control circuits:										
10.3.1. Controls Physics							**K**	**K**	**K**	**K**
10.3.1.1. Boolean Math Logic							**K**	**PK**	**PK**	**PK**
10.3.1.2. Truth Table							**K**	**PK**	**PK**	**PK**
10.3.1.3. Logic Circuits							**K**	**PK**	**PK**	**PK**
10.3.1.4. Component Timing							**K**	**PK**	**PK**	**PK**
10.3.1.5. Microprocessors/PLC's,							**K**	**PK**	**PK**	**PK**
10.3.1.6. Amplified Control Circuits							**K**	**PK**	**PK**	**PK**
10.4. System/Subsystem control strategies							**K**	**K**	**K**	**K**
10.5. Perform system changes:										
10.5.1. Programing							**K**	**PK**	**PK**	**PK**
10.5.2. Motion and timing							**K**	**PK**	**PK**	**PK**
10.5.3. Loop Tuning							**K**	**PK**	**PK**	**PK**
10.6. Energy monitoring & control systems (EMCS) operating principles and components:										
10.6.1. Energy Reports							**K**	**K**	**K**	**K**
10.6.2. Trending							**K**	**PK**	**PK**	**PK**
10.6.3. Alarming							**K**	**PK**	**PK**	**PK**
10.6.4. Reporting/Metering							**K**	**PK**	**PK**	**PK**

10.7. Computer Systems:										
10.7.1. Administrative Privileges							**K**	**PK**	**PK**	**PK**
10.7.2. Network Security								**PK**	**PK**	**PK**
10.7.3. Encryption Standards								**K**	**K**	**K**
10.7.4. Network Topologies								**K**	**K**	**K**
10.7.5. Principles of Network Distribution Cabling								**K**	**K**	**K**
10.7.6. Software Programing							**K**	**PK**	**PK**	**PK**
10.7.7. Communications:										
10.7.7.1. BACnet, Modbus, LonWorks							**K**	**K**	**K**	**K**
10.7.7.2. Wireless							**K**	**K**	**K**	**K**
10.7.7.3. Point/Multi Point to Point							**K**	**K**	**K**	**K**
10.8. Install							**K**	**PK**	**PK**	**PK**
10.9. Troubleshoot and repair							**K**	**PK**	**PK**	**PK**
10.10. Perform Preventative Maintenance							**P**	**PK**	**PK**	**PK**
11. HEATING & HOT WATER SYSTEMS: TR: ASHRAE Std 14										
11.1. Characteristics of fuels (oil, gas)							**K**	**K**	**K**	**K**
11.2. Types and principles of fuel systems							**K**	**K**	**K**	**K**
11.3. Environmental concerns							**K**	**K**	**K**	**K**
11.4. Inspect systems for leaks							**P**	**PK**	**PK**	**PK**
11.5. Install and maintain fuel lines							**K**	**PK**	**PK**	**PK**
12. BURNERS:										
12.1. Construction features and operation of burners (oil and gas)							**K**	**K**	**K**	**K**
12.2. Properties of combustion							**K**	**K**	**K**	**K**
12.3. Combustion analyzers							**K**	**K**	**K**	**K**
12.4. Perform pre-operational inspections							**K**	**PK**	**PK**	**PK**
12.5. Perform operational test							**K**	**PK**	**PK**	**PK**
12.6. Perform combustion analysis							**K**	**PK**	**PK**	**PK**
12.7. Compute combustion efficiency							**K**	**PK**	**PK**	**PK**
12.8. Adjust fuel/air ratio for proper combustion efficiency							**K**	**PK**	**PK**	**PK**
12.9. Perform preventative maintenance							**P**	**PK**	**PK**	**PK**

12.10. Repair components and compliance:										
12.10.1. Troubleshoot							**K**	**PK**	**PK**	**PK**
12.10.2. Correct malfunctions							**K**	**PK**	**PK**	**PK**
12.10.3. Environmental requirements							**K**	**K**	**K**	**K**
12.10.4. Electronic linkage-less burner control:										
12.10.4.1. Install								**PK**	**PK**	**PK**
12.10.4.2. Set-up								**PK**	**PK**	**PK**
12.10.4.3. Troubleshoot								**PK**	**PK**	**PK**
12.10.4.4. Adjust efficiency								**PK**	**PK**	**PK**
13. FORCED AIR AND RADIANT HEATING:										
13.1. Types and operation							**K**	**K**	**K**	**K**
13.2. Perform Pre-operational Inspection							**K**	**PK**	**PK**	**PK**
13.3. Perform operational test							**K**	**PK**	**PK**	**PK**
13.4. Preventive maintenance							**P**	**PK**	**PK**	**PK**
13.5. Troubleshoot and repair							**K**	**PK**	**PK**	**PK**
13.6. High efficiency condensing furnace:										
13.6.1. Install							**P**	**PK**	**PK**	**PK**
13.6.2. Troubleshoot and repair							**K**	**PK**	**PK**	**PK**
13.6.3. Perform preventive maintenance							**P**	**PK**	**PK**	**PK**
13.7. Portable Oil-Fired Heaters:										
13.7.1. Set-up							**K**	**PK**	**PK**	**PK**
13.7.2. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
13.7.3. Perform preventative maintenance							**K**	**PK**	**PK**	**PK**
13.8. Radiant Heating:										
13.8.1. Types							**K**	**K**	**K**	**K**
13.8.2. Operation							**K**	**K**	**K**	**K**
13.8.3. Perform pre-operational inspection							**K**	**PK**	**PK**	**PK**
13.8.4. Perform operational test							**K**	**PK**	**PK**	**PK**
13.8.5. Troubleshoot and repair							**K**	**PK**	**PK**	**PK**
13.8.6. Perform preventive maintenance							**P**	**PK**	**PK**	**PK**

14. STEAM GENERATION:										
14.1. Properties of steam							**K**	**K**	**K**	**K**
14.2. Principles of distribution systems (types and components)							**K**	**K**	**K**	**K**
14.2.1. Steam trap types and application							**K**	**K**	**K**	**K**
14.2.2. Steam heat exchangers							**K**	**K**	**K**	**K**
14.2.3. Condensate return systems							**K**	**K**	**K**	**K**
14.3. Principles of boilers (fire tubes, water tubes, and cast iron)							**K**	**K**	**K**	**K**
14.4. Construction features of boilers (waterside, fireside, and external fittings)							**K**	**K**	**K**	**K**
14.5. Auxiliary equipment (feedwater system, safety devices, and controls)							**K**	**K**	**K**	**K**
14.6. Perform pre-operational inspection							**K**	**PK**	**PK**	**PK**
14.7. Perform operational test							**K**	**PK**	**PK**	**PK**
14.8. Perform preventive maintenance							**K**	**PK**	**PK**	**PK**
14.9. Repair boilers and/or auxiliary equipment:										
14.9.1. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
14.9.2. Replace/remove boiler sections							**K**	**PK**	**PK**	**PK**
14.10. Prepare boiler for inspection (Types A, B, C, D)							**K**	**PK**	**PK**	**PK**
14.11. High efficiency condensing boilers:										
14.11.1. Function and maintenance of flue gas condensate neutralizers							**K**	**PK**	**PK**	**PK**
14.11.2. Install							**K**	**PK**	**PK**	**PK**
14.11.3. Troubleshoot and repair								**P**	**PK**	**PK**
14.11.4. Perform preventative maintenance							**P**	**PK**	**PK**	**PK**
14.12. In-floor heating systems:										
14.12.1. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
14.13. Sidewalk heating systems:										
14.13.1. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
15. AIR CONDITIONING & REFRIGERATION SYSTEMS: **TR: Fundamentals of HVAC/R 2nd Ed AHRI; Modern Refrigeration and Air Conditioning, 19th Ed**										
15.1. Refrigerants (types and uses)							**K**	**K**	**K**	**K**
15.2. Oils (types and uses)							**K**	**K**	**K**	**K**
15.3. Universal Certification							**K**	**PK**	**PK**	**PK**

15.4. Basic Refrigeration:										
15.4.1. Refrigeration cycle							**K**	**K**	**K**	**K**
15.4.2. Locate refrigerant leaks							**K**	**PK**	**PK**	**PK**
15.4.3. Recover and recycle refrigerant							**K**	**PK**	**PK**	**PK**
15.4.4. Pump down system							**K**	**PK**	**PK**	**PK**
15.4.5. Pressure check system							**K**	**PK**	**PK**	**PK**
15.4.6. Charge system:										
15.4.6.1. Determine and weigh in correct charge							**K**	**PK**	**PK**	**PK**
15.4.6.2. Compressor burnout procedures							**K**	**K**	**K**	**K**
15.4.6.3. Calculate and adjust superheat							**K**	**PK**	**PK**	**PK**
15.4.6.4. Calculate and adjust sub-cooling							**K**	**PK**	**PK**	**PK**
15.4.6.5. Utilize pressure-temperature charts							**P**	**PK**	**PK**	**PK**
15.4.6.6. Plot pressure enthalpy charts								**PK**	**PK**	**PK**
15.5. Tools:										
15.5.1. Leak detectors:										
15.5.1.1. Soap bubbles							**K**	**K**	**K**	**K**
15.5.1.2. Fluorescent Dye							**K**	**K**	**K**	**K**
15.5.1.3. Ultrasonic							**K**	**K**	**K**	**K**
15.5.1.4. Electronic							**K**	**K**	**K**	**K**
15.5.2. Recovery machines							**K**	**K**	**K**	**K**
15.5.3. Refrigerant cylinders (types, sizes, fill capacities)							**K**	**K**	**K**	**K**
15.5.4. Vacuum pump							**K**	**K**	**K**	**K**
15.5.5. Refrigerant Scales							**K**	**K**	**K**	**K**
15.5.6. Refrigerant gauges:										
15.5.6.1. Analog							**K**	**K**	**K**	**K**
15.5.6.2. Digital							**K**	**K**	**K**	**K**
15.5.6.3. Bluetooth							**K**	**K**	**K**	**K**
15.5.7. Maintain and use equipment							**P**	**PK**	**PK**	**PK**
15.5.8. APIMS required paperwork (Refrigerant tracking)							**K**	**PK**	**PK**	**PK**

15.6. Refrigeration components:										
15.6.1. Compressor types:										
15.6.1.1. Reciprocating							**K**	**K**	**K**	**K**
15.6.1.2. Centrifugal							**K**	**K**	**K**	**K**
15.6.1.3. Rotary							**K**	**K**	**K**	**K**
15.6.1.4. Screw							**K**	**K**	**K**	**K**
15.6.1.5. Scroll							**K**	**K**	**K**	**K**
15.6.2. Evaporators:										
15.6.2.1. Direct expansion							**K**	**K**	**K**	**K**
15.6.2.2. Indirect expansion							**K**	**K**	**K**	**K**
15.6.3. Condensers:										
15.6.3.1. Air cooled							**K**	**K**	**K**	**K**
15.6.3.1.1. Receivers							**K**	**K**	**K**	**K**
15.6.3.1.2. Condenser pressure regulator (CPR) (Head-Master)							**K**	**PK**	**PK**	**PK**
15.6.3.1.3. Fan cycling controls							**K**	**PK**	**PK**	**PK**
15.6.3.2. Water cooled/cooling tower							**K**	**K**	**K**	**K**
15.6.4. Metering devices:										
15.6.4.1. Fixed metering							**K**	**K**	**K**	**K**
15.6.4.2. Thermostatic expansion valve							**K**	**K**	**K**	**K**
15.6.4.3. Electronic expansion valve							**K**	**K**	**K**	**K**
15.6.5. Install							**K**	**PK**	**PK**	**PK**
15.6.7. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
15.7. Refrigeration systems:										
15.7.1. Commercial:										
15.7.1.1. Walk in freezers							**K**	**PK**	**PK**	**PK**
15.7.1.2. Walk in coolers							**K**	**PK**	**PK**	**PK**
15.7.2. Ice machines							**K**	**PK**	**PK**	**PK**
15.7.3. Cold storage (permanent/portable)							**K**	**PK**	**PK**	**PK**
15.7.4. Install							**K**	**PK**	**PK**	**PK**
15.7.5. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
15.7.6. Perform preventative maintenance							**P**	**PK**	**PK**	**PK**

15.8. Refrigeration Controls:										
15.8.1. High pressure & low-pressure switches							**K**	**PK**	**PK**	**PK**
15.8.2. Thermostats (mechanical or digital)							**K**	**PK**	**PK**	**PK**
15.8.3. Oil pressure safety							**K**	**PK**	**PK**	**PK**
15.8.4. Condenser fan variable speed drives (VFD)							**K**	**PK**	**PK**	**PK**
15.9. Capacity Controls:										
15.9.1. Unloaders							**K**	**PK**	**PK**	**PK**
15.9.2. Hot Gas By-Pass Valves							**K**	**PK**	**PK**	**PK**
15.10. Air conditioning systems:										
15.10.1. Package units							**K**	**K**	**K**	**K**
15.10.2. Split systems							**K**	**K**	**K**	**K**
15.10.3. Heat pumps:										
15.10.3.1. Air to Air							**K**	**K**	**K**	**K**
15.10.3.2.Water to water							**K**	**K**	**K**	**K**
15.10.3.3. Water to air							**K**	**K**	**K**	**K**
15.10.3.4. Geothermal wells vertical & horizontal							**K**	**K**	**K**	**K**
15.10.4. Variable Refrigerant Flow (VRF)							**K**	**K**	**K**	**K**
15.10.5. Install								**P**	**PK**	**PK**
15.10.6. Troubleshoot and Repair							**P**	**PK**	**PK**	**PK**
15.10.7. Perform preventative maintenance							**P**	**PK**	**PK**	**PK**
15.11. Equipment cooling:										
15.11.1 Computer room air conditioners:										
15.11.1.1. Direct expansion (CRAC)							**K**	**K**	**K**	**K**
15.11.1.2. Chilled water (CRAH)							**K**	**K**	**K**	**K**
15.11.2. Humidifiers							**K**	**K**	**K**	**K**
15.11.3. Mini-split system							**K**	**K**	**K**	**K**
15.11.4. Install								**PK**	**PK**	**PK**
15.11.5. Troubleshoot and repair								**P**	**PK**	**PK**
15.11.6. Perform preventative maintenance							**PK**	**PK**	**PK**	**PK**

15.12. Chillers:										
15.12.1. Types:										
15.12.1.1. Air cooled							**K**	**K**	**K**	**K**
15.12.1.2. Water cooled							**K**	**K**	**K**	**K**
15.12.1.3. Evaporative condensed chillers							**K**	**K**	**K**	**K**
15.12.2. Sub-categories of chillers:										
15.12.2.1. Reciprocating							**K**	**K**	**K**	**K**
15.12.2.2. Centrifugal							**K**	**K**	**K**	**K**
15.12.2.3. Screw							**K**	**K**	**K**	**K**
15.12.2.4. Scroll							**K**	**K**	**K**	**K**
15.12.2.5. Absorption							**K**	**K**	**K**	**K**
15.12.3. Install								**PK**	**PK**	**PK**
15.12.4. Troubleshoot and repair								**P**	**PK**	**PK**
15.12.5. Perform preventative maintenance							**P**	**PK**	**PK**	**PK**
16. AIR COMPRESSING EQUIPMENT (non-aircraft generation/non-breathable): TR: AFI 32-1068)										
16.1. Types of compressors										
16.1.1. Reciprocating							**K**	**K**	**K**	**K**
16.1.2. Rotary screws							**K**	**K**	**K**	**K**
16.1.3. Centrifugal							**K**	**K**	**K**	**K**
16.2. Auxiliary equipment:										
16.2.1. Aftercoolers							**K**	**K**	**K**	**K**
16.2.2. Filters							**K**	**K**	**K**	**K**
16.2.3. Separators							**K**	**K**	**K**	**K**
16.2.4. Dryers							**K**	**K**	**K**	**K**
16.3. Install							**K**	**PK**	**PK**	**PK**
16.4. Troubleshoot and repair							**K**	**P**	**PK**	**PK**
16.5. Perform preventative maintenance							**K**	**PK**	**PK**	**PK**
17. WATER TESTING & TREATMENT: TR: AFI 32-1054; BETZ Handbook of Industrial Water Conditioning; Principles of Industrial Water Treatment 9th Ed										
17.1. Water characteristics							**K**	**K**	**K**	**K**
17.2. Purpose and types of tests							**K**	**K**	**K**	**K**
17.3. Internal treatment methods:										
17.3.1. Determine and adjust water levels								**P**	**PK**	**PK**
17.3.2. Determine and adjust treatment levels								**P**	**PK**	**PK**
17.3.3. Determine and adjust conductivity								**P**	**PK**	**PK**

17.4. External treatment methods:										
17.4.1. Chemical feeding equipment/condensate return								**K**	**K**	**K**
17.4.2. Environmental requirements								**K**	**K**	**K**
17.4.3. Glycol characteristics:										
17.4.3.1. Chemical constituents (pH, freeze point, additives, etc.)								**K**	**K**	**K**
17.4.4. Perform water treatment								**P**	**PK**	**PK**
18. WORKCENTER MANAGEMENT:										
18.1. NEXGEN-TRIRIGA:										
18.1.1. Input labor								**P**	**PK**	**PK**
18.1.2. Create purchase request								**P**	**PK**	**PK**
18.1.3. Manage Preventative Maintenance (PM) program								**P**	**PK**	**PK**
18.1.4. Schedule work tasks									**PK**	**PK**

Attachment 3 - Air Force Qualification Training Package (AFQTP) Tracker

A3. AFQTP Documentation Record

A3.1. To ensure each HVAC/R Specialist is trained to the correct standard an AFQTP has been developed for each task group identified in their STS. These AFQTPs are to be used by the employee, trainer and certifier in their on-the-job training program for qualification training and developmental training.

A3.2. These AFQTPs ensure all aspects of the task are covered sufficiently and provide additional task knowledge, in preparation for hands-on training. AFQTPs summarize procedures on a task performance checklist for use by trainers, certifiers, and employees.

A3.2.1. The UTM or supervisor can download paper based AFQTP's. Paper-based AFQTP's are found on CE DASH under documents in the AFQTP folder.

A3.2.2. In addition to the paper-based AFQTPs there are web-based courses or assessments developed for certain tasks that are available on myLearning under AFCEC in the Home Station topic area.

A3.3. Documentation. Before a core or diamond task can be signed off in the JQS section of the individual automated training record, the task must be signed off in the QTP section first.

A3.4. AFQTP Tracker

Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
	Core	Deployment	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
SAFETY						
Demonstrate the safety practices associated when performing PM on HVAC equipment, components, and accessories **(P)**						
Demonstrate the safety precautions associated with troubleshooting cooling systems **(P)**						
Perform "Lock out and Tag" procedures on cooling equipment **(P)**						
Perform first aid procedures for electrical shock **(P)**						
Identify proper safety procedures to follow on discovery of a gas leak **(K)**						
Identify relevant safety topics and brief shop personnel to aid in the development of a culture of safety **(P)**						
AIR DISTRIBUTION & VENTILATION						
Perform preventative maintenance on an Air Handling Unit (AHU) using an AF PMTL **(P)**						
Replace and adjust a sheave / pulley on a belt driven blower **(P)**						
Identify & interpret a building indoor air quality (IAQ) inspection/evaluation **(P)**						
Calculate duct size, using duct sizing formulas for ductwork to be used in a temporary shelter (e.g., KSPAN, LAMS, B-Hut) **(P)**						
HYDRONICS						
Remove, install, and align centrifugal pumps & various types of pump couplings & seals **(P)**						
Identify, explain, remove, and install various types of gaskets and packing **(P)**						
Explain cavitation and demonstrate how to remove air from a hydronic system **(P)**						
Remove, fabricate, and install black iron pipe and fittings **(P)**						
ELECTRICAL						
Perform electrical tests and troubleshooting of compressors and motors (1 & 3 Phase) **(P)**						
Describe the operation and characteristics of an Electronically Commutated Motor) **(K)**						
Install different types of motors (e.g., Shaded pole, split phase, PSC, Capacitor start capacitor run, and ECM) (3ph wire for Wye & Delta) **(P)**						
Isolate and correct malfunctions on a Chiller **(P)**						
Isolate and correct malfunctions on a hot water and steam boiler **(P)**						
Isolate and correct malfunctions on a Packaged AC unit **(P)**						

Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
	Core	Deployment	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
Isolate and correct malfunctions on a Walk in Refrigerator/Freezer **(P)**						
CONTROL SYSTEMS (CS)						
WENG 170 - Cybersecurity for Control Systems Course WBT **(K)**						
Demonstrate how to install typical sensors, actuators, power wiring, and communication wiring **(P)**						
Describe and test thermistor type & temperature sensors (PTC & NTC) **(P)**						
Program and operate a basic direct digital controller **(P)**						
Adjust Building Automation System parameters and explain how to verify the change **(P)**						
Perform troubleshooting on Building Automation System and correct system malfunctions **(P)**						
Evaluate and Install a Variable Speed Drive (VSD) & Variable Frequency Drive (VFD) **(P)**						
HEATING						
Clean, start up, and shut down a hot water Boiler **(P)**						
Adjust various types of igniters (e.g., Pilot Burner & Electric Spark) **(P)**						
Complete HVAC MEET 2 Course (Dominate in Cold/Artic Environment) (Contingency) **(P)**						
COOLING & REFRIGERATION						
Charge refrigerant into a system by the following methods: Weight, Subcooling, Superheat, Charging pressure chart **(P)**						
Demonstrate proper soldering and brazing techniques **(P)**						
Remove and install a compressor (5 ton or less system) **(P)**						
Evacuate a system using a vacuum pump and micron gauge, and explain deep vacuum and triple evacuation method **(P)**						
Use equipment manufacturer's troubleshooting aids to troubleshoot HVAC system components and accessories **(P)**						
Use the manufacturer's literature to perform preventive maintenance **(P)**						
Safely join tubing by using flare, swag and compression fittings and identify the proper torque setting **(P)**						
Correctly cut and bend copper tubing **(P)**						
Isolate and correct mechanical malfunctions in a Cooling and a Refrigeration system (e.g., Coil leaks, oil issues, faulty boards, over/under charge) **(P)**						

Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
	Core	Deployment	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
Complete HVAC MEET 1 Course (Dominate in Hot/Desert Environment) **(P)**						
Install a Commercial Packaged air conditioning unit **(P)**						
Perform a contingency load estimate to determine the heating and cooling load of temporary shelters (e.g., KSPAN, LAMS, B-Hut) **(P)**						
PROGRAMS						
Assist shop APIMS program manager **(P)**						
WORK CENTER MANAGEMENT						
Identify and balance HVAC Preventive Maintenance requirements (i.e., manpower, equipment, materials, etc.) **(P)**						
Complete AFIT WMGT 301 Intro to Asset Management **(P)**						
Complete NexGen IT Training **(P)**						

Appendix A - Acronyms

Acronym	Definition
AETC	Air Education and Training Command
AETCI	Air Education and Training Command Instruction
AF	Air Force
AFCCFM	Air Force Civilian Career Field Manager
AFCDA	Air Force Career Development Academy
AFCEC	Air Force Civil Engineer Center
AFCEC/COF	Air Force Civil Engineer Center Force Development Division
AFIT	Air Force Institute of Technology
AFQTP	Air Force Qualification Training Package
AFSC	Air Force Specialty Code
ALS	Airman Leadership School
CBB	Career Building Block
CBT	Computer-Based Training
CDC	Career Development Course
CE	Civil Engineer
CFETP	Career Field Education and Training Plan
CFM	Career Field Management
COTS	Commercial Off The Shelf
DAFI	Department of Air Force Instruction
DAFMAN	Department of Air Force Manual
DL	Distance Learning
DoD	Department of Defense
EPME	Enlisted Professional Military Education
FSTS	Family Series Training Standard
FWS	Federal Wage System
HAF	Headquarters Air Force
HQ	Headquarters
IAW	In Accordance With
JQS	Job Qualification Standard
MTT	Mobile Training Team
NCOA	Noncommissioned Officer Academy
OJT	On-the-Job Training
PCs	Progress Checks
PME	Professional Military Education
SCPD	Standard Core Personnel Document
SEI	Special Experience Identifier
SNCOA	Senior Noncommissioned Officer Academy
SOS	Squadron Officers School
STS	Specialty Training Standard
TOC	Table of Contents
TR	Technical References
USAF	United States Air Force

Acronym	Definition
UETM	Unit Education Training Manager
UTM	Unit Training Manager
WBT	Web-Based Training

Appendix B - Master List of Links

Content	Location	Web Address
AETCI 36 - 2651, *Basic Military and Technical Training*	Part II, B4	https://static.e-publishing.af.mil/production/1/aetc/publication/aetci36-2651/aetci36-2651.pdf
AFIT Course List	Appendix C	https://www.afit.edu/ce/
Air Force Portal	Part I, B3 Part I, B3.1	https://www.my.af.mil/
Air University URL	Part I, B3.2	https://www.airuniversity.af.edu/GCPME/
CE Career Field Team SharePoint	Appendix C	https://usaf.dps.mil/sites/afpc-home/DPZ/DPZC/DPZCD/SitePages/Home.aspx?e=1:c6262c52de91498ca691729657201814
CE DASH	Attachment 3, A3.2.1	https://usaf.dps.mil/teams/10758/CEtraining/SitePages/Home.aspx
CE Training Email	Part I, A4	afcec.ce.training@us.af.mil
Classification Standards	Appendix C	https://www.opm.gov/policy-data-oversight/classification-qualifications/classifying-federal-wage-system-positions/
DAFI 36-2670, *Total Force Development*	Part I, B4 Part I, B4.1 Part II, A2 Part II, A2.6	https://static.e-publishing.af.mil/production/1/af_a1/publication/dafi36-2670/dafi36-2670.pdf
DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development*	Part II, A2 Part II, A2.6	https://static.e-publishing.af.mil/production/1/af_a4/publication/dafman32-2670/dafman32-2670_.pdf
DoD Civilian COOL	Appendix C	http://dod-civ-cool-review.s3.amazonaws.com/index.htm
Education and Training Course Announcements	Part II, D1 Appendix C	https://usaf.dps.mil/teams/app10-etca/SitePages/home.aspx?isdlg=1&IsDlg=1
Enlisted Professional Military Education (EPME)	Appendix C	https://www.airuniversity.af.edu/
Federal Building Competency Model	Attachment 2, A.2.2	https://www.gsa.gov/system/files/FBPTA%20Competencies%20Model%20update%20DEC%202024.xlsx
Job Series CBBs	Part I, B5.1.1	https://usaf.dps.mil/sites/afpc-home/DPZ/DPZC/DPZCD/Shared%20Documents/Forms/AllItems.aspx?csf=1&web=1&e=aGJrkM&CID=c8d15d94%2D9aea%2D4d66%2Dad5e%2D180200c4b44a&FolderCTID=0x012000EF779A2927E84044A72F4C3AF25B1CD2&id=%2Fsites%2Fafpc%2Dhome%2FDPZ%2FDPZC%2FDPZCD%2FShared%20Documents%2FCE%20Career%20Building%20Blocks
Civilian Force Development Homepage	Part II, D5 Appendix C	https://myfss.us.af.mil/USAFCommunity/s/knowledge-detail?xid=13085
myLearning	Part II, A2.6 Part II, C1.2.1 Part II, C1.2.2 Part II, C2.1 Attachment 3, A3.2.2 Appendix C	https://lms-jets.cce.af.mil/moodle/

Content	Location	Web Address
Red Vector	Appendix C	https://afcec.redvector.com/lpe/course/search/b2b
FWS Series Qualification Standard (WGSQS)	Appendix C	https://www.opm.gov/policy-data-oversight/classification-qualifications/federal-wage-system-qualifications/

Appendix C - Resources

Resource/Term	Definition
Air Force Civil Engineer Center (AFCEC)	The focal point for all Civil Engineer training development.
Air Force Institute of Technology (AFIT)	Provides vital, relevant, and connected education that enables Airmen to be ready engineers and great leaders who know how to build sustainable installations to last while leading the change for the Civil Engineer career field. Review the Course list.
Air Force Qualification Training Package (AFQTP)	An instructional package designed for use as a training resource to qualify, or aid qualification, in a duty position or program, or on a piece of equipment. AFQTPs identify the AF's standardized method for performing the task. The AFQTP may be printed (paper-based), computer-based, in other audiovisual media formats, or all three.
Career Development Course (CDC)	A self-paced correspondence course published to provide the information necessary to satisfy the career knowledge component of OJT. These courses are developed from references identified in the CFETP. CDCs will contain information on basic principles, techniques, and procedures common to a military AFSC or civilian job series. They do not contain information on specific equipment or tasks unless best illustrating a procedure or technique having utility to the entire career field.
Career Field Education and Training Plan (CFETP)	A comprehensive, multipurpose document encapsulating the entire spectrum of education and training for various FWS series. It outlines a logical growth plan that includes training resources and is designed to make job series training identifiable, to eliminate duplication, and to ensure this training is budget defensible.
Career Field Manager (CFM)	An individual on the Air Staff charged with the responsibility for overseeing all training and career field management aspects of multiple AF job series in a functional area.
CE Career Field Team SharePoint	Contains information on leadership courses and civilian tuition assistance for degree programs. See: Civil Engineer Career Field Team SharePoint.
Civilian Force Development Homepage	Contains descriptions and requirements for several civilian training opportunities such as degree programs and civilian leadership courses. Click here to view the homepage.
Classification Standard	Job grading standards provide information used in determining the occupational series and title of jobs performing trades, craft, and labor work in the Federal Government. They also provide grading criteria for positions classified under the Federal Wage System (FWS). Click here to view Classification Standards.

Commercial Off the Shelf (COTS)	Commercially procured training products or in-resident vendor training.
Computer-Based Training (CBT)	A self-paced stand-alone computer product used to deliver interactive subject and task knowledge.

Resource/Term	Definition
Core Tasks	Mandatory tasks which the CFM has identified as a minimum qualification requirement within a job series or duty position. These tasks are derived from the Standard Core Personnel Document (SCPD) for each job series.
Distance Learning (DL)	Includes Video Tele-seminar (VTS), Video Tele-training (VTT), and CBT. Formal courses that a training wing or a contractor develops for export to a field location (in place of resident training) for trainees to complete without the on-site support of the formal school instructor. For instance, courses are offered by AFIT, Air University, and Training Detachments.
DoD Civilian COOL	On-line source for civilian credentialing opportunities.
Duty Position Tasks	Tasks identified by the work center supervisor as critical and common training tasks needed for the duty position and mission accomplishment.
Education & Training Course Announcements (ETCA)	Web platform, which contains descriptions, requirements, and reporting procedures for in resident Air Force courses.
Enlisted Professional Military Education (EPME)	EPME provides a continuum of learning through progressive courses concentrated on developing Military and Civilian Airmen. EPME plays a vital role in preparing Civilian Airmen for increased supervision, leadership, and management challenges. The three levels of AF EPME are Airman Leadership School, Noncommissioned Officer Academy and Air Force Senior Noncommissioned Officer Academy. All levels of EPME are available to FWS civilians in accordance with DAFI 36-2685 and DAFI 36-2687. Click here for more Information about EPME.
Family Series Training Standard (FSTS)	Describes skills and knowledge that FWS Employees in a particular job series need on the job and for future career development opportunities. It further serves as the overall training requirements for a Wage Series taught in the resident and nonresident courses as well as an on-the-job training program
Functional Advisory Council (FAC)/Federal Wage System Panel	The Federal Wage System Panel is one of the four panels that make up the Civil Engineer FAC. The Federal Wage System Panel charter is to work issues, develop policy, and provide recommendations to the FAC on matters related to civilian FWS requirements. The Federal Wage System Panel works through the FAC, in service to the CE Total Force community.
myLearning	Anytime, anywhere learning within DoD consisting of instructional modules comprised of sharable content objectives in an Internet/Intranet environment. Click here for mylearning.
On-the-Job Training (OJT)	Hands-on, over-the-shoulder training conducted to certify personnel in job qualification (duty position certification) training.
Proficiency Training	Additional training, either in-residence, advanced/supplemental training courses, or on-the-job training provided to personnel to increase their skills and knowledge beyond the minimum.

Resource/Term	Definition
Red Vector	Commercial web-based training available free of charge to CE employees. Courses are effective for certification renewal and earn Continuing Educations Units (CEUs). Click here for Red Vector site.
Regional Training Site (RTS)	Total Force training centers managed by the Air National Guard. These sites offer training on specialized military equipment and are available to civilians who require training for local mission needs. Training can be coordinated through the FWS Force Development Team at AFCEC/COF.
Resource Constraints	Resource deficiencies, such as money, facilities, time, manpower, or equipment that precludes desired training from being delivered.
Total Force	All collective AF components (Active Duty, Reserve, Guard, and Civilian elements) of the USAF.
Vendor Training	Training provided to the Air Force by a third party. Typically, a private vendor in the private sector not affiliated with the Department of Defense. Vendor training can be in the form of Web-Based Training, Computer-Based Training, or in resident training at a temporary duty location or even hosted on an Air Force installation.
Federal Wage System Series Qualification Standard (WGSQS)	A comprehensive task list that describes a particular series or duty position. Used by supervisors to document task qualifications. The tasks on the AFWGSQS are common to all persons serving in the described duty position. Click here for qualifications.
FWS Series Training	A mix of formal training (technical school) and informal training (on-the-job) to develop, maintain and enhance FWS series specific technical skills.
Web-Based Training	A form of Distance Learning. The term Web-Based just means the training is online and requires access to the internet in addition to the actual course content.